Shakil Ahmed

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Research Interest

Prospective PhD candidate in deep learning and computer vision with peer-reviewed publications. Skilled in Python, PyTorch, CNNs, Transformers, foundation models, and domain adaptation, with research interest in AI-based biodiversity monitoring and edge computing.

Education

Aarhus University — Aarhus, Denmark
Master of Science in Computer Science — *Sept 2024 – Current*

- **Relevant Coursework:** Deep Learning for Visual Recognition, Data Mining, Data Visualization, Statistical and Machine Learning in Bioinformatics.
- **Tentative Thesis:** Tackling the Annotation Bottleneck: Unsupervised Fine-Grained Clustering of Insect Biodiversity using Vision Foundation Models.
- **Supervisors:** Kim Bjerge and Christian Nørgaard Storm Pedersen.

American International University-Bangladesh — Dhaka, Bangladesh
Master of Science in Computer Science — *2021*

- **CGPA:** 4.00/4.00 (*Summa Cum Laude*)
- **Thesis:** Attention-Based Bi-LSTM and Sequential LSTM for Describing Video in the Textual Formation.(**Published**)
- **Relevant Coursework:** Computer Vision and Pattern Recognition, Data Mining and Warehousing, Image Processing, Natural Language Processing, Human Computer Interaction.

American International University-Bangladesh — Dhaka, Bangladesh
Bachelor of Science in Computer Science — *2020*

- **CGPA:** 3.86/4.00 (*Magna Cum Laude*)
- **Thesis:** An Observation of Vision Based Underwater Object Detection and Tracking. (**Published**)
- **Relevant Coursework:** Computer Vision and Pattern Recognition, Data Mining and Warehousing, Image Processing, Natural Language Processing, Human Computer Interaction.

Selected Publications

Journal Articles

- **Att-BiL-SL: Attention-Based Bi-LSTM and Sequential LSTM for Describing Video in the Textual Formation. Ahmed, S.**; Saif, A.F.M.S.; Hanif, M.I.; Shakil, M.M.N.; Jaman, M.M.; Haque, M.M.U. et. al. (2021) Applied Sciences, 12(1), 317. doi:10.3390/app12010317.
- **Anti-molestation: An IoT-based device for women's self-security system to avoid unlawful activities.** Hanif, M. I.; **Ahmed, S.**; Akanda, W.; and Barman, S. (2020) International Journal of Advanced Computer Science and Applications, 11(11). doi:10.14569/IJACSA.2020.0111188.

Conferences

- **Multicriteria Decision Analysis for Optimal Internet Service Provider Selection Using Calibrated Random Forest.** Haque, M.M.U.; **Ahmed, S.**; Aftab, R. S.; Miah, M. S. U.; Akanda, W.; Bhowmik, A. 2024 Asian Conference on Communication and Networks (ASIANComNet), Bangkok, Thailand, 2024, pp. 1–6. doi:10.1109/ASIANComNet63184.2024.10811091.
- **An observation of vision-based underwater object detection and tracking. Ahmed, S.**; Khan, M. F. R.; Labib, M. F. A.; and Chowdhury, A. E. 2020 3rd International Conference on Emerging Technologies in Computer Engineering: Machine Learning and Internet of Things (ICETCE), pp. 117–122, Jaipur, India. doi:10.1109/ICETCE48199.2020.9091752.

Experience

Aarhus University Sept 2025 – Current
Teaching Assistant *Deep Learning for Visual Recognition*

- Facilitated end-to-end implementation of State-of-the-Art (SOTA) vision models, guiding 50+ students through Hyperparameter Tuning, Loss Function Optimization, and Model Benchmarking using PyTorch and Tensorflow.

Aarhus University Feb 2025 – May 2025
Teaching Assistant *Computer Architecture, Network and Operating Systems*

- Formulated solutions for low-level systems problems and evaluate student assignments.
- Provided technical support for students in the Lab and Study Cafe environments.

IT Grarage Mar 2022 – Jul 2024
Mobile Application Developer *Flutter*

- Architected and deployed robust cross-platform applications for government-based projects.
- Collaborated with cross-functional teams to build and deploy Android/iOS apps to global stores.

Ministry of ICT - Cross-platform Mobile Application Development - Under ICT Division Sept 2021 – Jan 2022
Lecturer and Trainer *Flutter*

- Supervised the development of 16 student-led mobile projects and taught Flutter/Dart frameworks.

American International University-Bangladesh Jun 2019 – Sept 2019
Intern *Object Oriented Programming 1 (JAVA)*

- Supervised group projects, assisted the faculty member in lab and helped students who needed extra support.

Projects

Deep Learning: Image Captioning with Resnet101 and LSTM with Attention Source Code

- Developed an Encoder-Decoder model using **ResNet-101** for feature extraction and **LSTM** for text generation.
- Implemented Soft-Attention Mechanism and Beam Search to visualize visual-semantic alignment, achieving **58.5 BLEU** score on the Flickr8k dataset.
- Visualized loss convergence and qualitative attention maps to validate visual-semantic alignment.

Data Visualization: Cell Tower Visualization Dashboard Source Code

- Engineered a high-performance dashboard for mapping global cell tower density using **Plotly and Dash**. Optimized the processing of large-scale datasets using Pandas to enable real-time spatial querying and visualization.

Technical Skills

Languages	:	Python, C++, Dart, R.
DL/CV	:	PyTorch, CNNs, Vision Transformers, RNNs, Attention Models, Transfer Learning, Data Augmentation, Domain Adaptation.
Data Science	:	NumPy, Pandas, Matplotlib, Plotly, Dataset preprocessing, model evaluation metrics.
Dev Tools	:	VS Code, Jupyter Notebook, Git, Play Store/App Store Deployment, Latex.

Awards

- 3 Dean List Honor - American International University-Bangladesh.
- CS FEST Runner-Up – American International University-Bangladesh.

Certifications

- Computer Vision Basics - Coursera (2020).
- Python for Everybody (5 Corses Specialization with Capstone Project) - Coursera (2020).

Academic Services and Voluntaries

- **Reviewer** - Neurocomputing; Image and Vision Computing; Journal of King Saud University: Computer and Information Sciences; Journal of Medical Engineering and Technology;